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Metabolic design in amino acid producing bacterium *Corynebacterium glutamicum*

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Abstract

The Gram-positive bacterium *Corynebacterium glutamicum* is used for the industrial production of amino acids, e.g. of L-glutamate and L-lysine. In the last 10 years, genetic engineering and amplification of relevant structural genes have become fascinating methods for the construction of strains with desired genotypes. By cloning and expressing the various genes of the L-lysine pathway in *C. glutamicum* we could demonstrate that an increase of the flux of L-aspartate semialdehyde to L-lysine could be obtained in strains with increased dehydrodipicolinate synthase activity. By combined overexpression of deregulated aspartate kinase and dihydrodipicolinate synthase, the L-lysine secretion could be increased (10–20%). Recently we detected that in *C. glutamicum* two pathways exist for the synthesis of DL-diaminopimelate and L-lysine. Mutants defective in one pathway are still able to synthesize enough L-lysine for growth, but the L-lysine secretion is reduced to 50–70%. Using NMR spectroscopy, we could calculate how much of the L-lysine secreted into the medium is synthesized via each pathway. Amplification of the feedback inhibition-insensitive homoserine dehydrogenase and homoserine kinase in a high L-lysine overproducing strain enabled channelling of the carbon flow from the intermediate aspartate semialdehyde towards homoserine, resulting in a high accumulation of L-threonine. For a further flux from L-threonine to L-isoleucine the allosteric control of threonine dehydratase must be eliminated. In addition to all steps considered so far to be important for amino acid overproduction, the secretion into the culture medium also has to be noted. Recently it could be demonstrated that L-glutamate, L-lysine and L-isoleucine are not secreted via passive diffusion but via specific active carrier systems. Analysis of lysine-overproducing *C. glutamicum* strains indicates that this secretion carrier has a strong influence on the overproduction of this amino acid. Thus, for the construction of strong amino acid overproducing strains by using the gene cloning techniques, the overexpression of the genes for the export systems also seems necessary.

Keywords: *Corynebacterium glutamicum*; Amino acid production; Metabolic design; L-Lysine; L-Threonine; L-Isoleucine

1. Introduction

L-Amino acids have a wide spectrum of commercial use as food additives, feed supplements, infusion compounds, therapeutic agents, and precursors for the synthesis of peptides or agrochemicals. In the mid-fifties, in Japan a bacterium was isolated which excreted large quantities of L-glutamic acid into the culture medium [1]. This bacterium, *Corynebacterium glutamicum*, is a short, aerobic, Gram-positive rod capable of growing on a simple medium. Under optimal conditions this organism converts

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about 100 g/l glucose into 50 g/l glutamic acid within a few days; about 350 000 t of L-glutamate are produced annually with this bacterium. In the last 30 years, various mutants of *C. glutamicum* were isolated which are also able to produce significant amounts of other L-amino acids. During the past decade, a new generation of strain improvements was developed using the gene cloning technique, as demonstrated by the following few examples.

2. L-Lysine producing strains

At present, L-lysine is produced in an amount of about 180000 t/a with strains of *C. glutamicum* or subspecies. The wild-types of these organisms do not secrete lysine, but strains producing this amino acid have been obtained by classical screening programs. During these procedures, mutants were selected primarily as resistant or sensitive to a variety of chemicals by assuming as targets enzymes of the amino acid pathway [2] or of central metabolism [3]. At each screening step, the mutant producing the highest titer of lysine was selected for subsequent mutagenesis. This empirical procedure resulted in strains with high productivity. However, their metabolic changes can be described neither qualitatively nor quantitatively. This also agrees with the fact that it is impossible to reconstruct a good lysine producer by using most of the published strain development schemes. The only exception is the use of the lysine analog *S*-2-aminoethyl-L-cysteine. Some of the mutants resistant to this analog secrete lysine, which can be attributed to an altered feedback response of the aspartate kinase [4]. This is the only enzyme involved in the lysine biosynthesis in *C. glutamicum* which is known to be controlled in its activity [5]. Aspartate kinase activity is strongly inhibited by the presence of lysine (1 mM) together with threonine (5 mM) (Fig. 1).

Recently, Cremer et al. [6] have described a rational approach to analysis of lysine production with *C. glutamicum* by overexpression of the various biosynthetic enzymes. Each of the six genes that are involved in the pathway of aspartate to lysine was overexpressed individually in the wild-type and a mutant with a feedback resistant aspartate kinase. Five- to ten-fold higher specific activities of each of the enzymes were observed in the recombinant *C. glutamicum* strains (Table 1). Analysis of lysine formation revealed that overexpression of the gene for the feedback-resistant aspartate kinase alone suffices to achieve lysine secretion in the wild-type. This result confirms that the deregulation of the aspartate kinase is the decisive first step in order to obtain *C. glutamicum* strains overproducing lysine.

Table 1
Enzyme activities [a] and lysine secretion of recombinants of *Corynebacterium glutamicum*

Plasmid	Relevant enzyme	Specific activity (U/mg) in strain			Mean (± SD) lysine [b] concentration (mM) with strain	
		13032	r-13032	r-52-5 [c]	r-13032	r-52-
pJC33	Aspartate kinase [d]	0.01	0.20	0.29	38 ± 3	48 ± 3
pJC31	Aspartate kinase, ß-subunit	–	0.02	–	24 ± 2	–
pJC300	Aspartate semialdehyde dehydrogenase	0.016	0.25	0.14	0 ± 1	38 ± 2
pJC24	Dihydrodipicolinate synthase	0.23	1.05	1.88	11 ± 2	48 ± 2
pJC25	Dihydrodipicolinate reductase	0.06	1.33	1.08	0 ± 1	41 ± 3
pJC40	Diaminopimelate dehydrogenase	0.13	2.5	2.1	0 ± 1	39 ± 2
pCT4-1	Diaminopimelate decarboxylase	0.03	0.28	0.27	0 ± 1	40 ± 3
pJC50	Aspartate kinase [d] plus	–	–	0.21	–	68 ± 5
	dihydrodipicolinate synthase	–	–	2.05	–	

[a] Cells were cultivated on minimal medium containing glucose and specific activities were determined after 48 h.
[b] Lysine was determined in supernatants after 48 h. Values are means from at least three independent cultivations. Strain 52-5 with no plasmid cultivated in parallel accumulated 40 ± 3 mM lysine.
[c] Strain 52-5 is a mutant with feedback-resistant aspartate kinase; r stands for the recombinant strain carrying one of the given plasmid.
[d] The specific kinase activities assayed in the presence of 60 mM lysine plus 60 mM threonine were 0.00 U/mg in 13032, 0.21 *U*/mg in r-13032, and 0.32 U/mg in r-52-5.

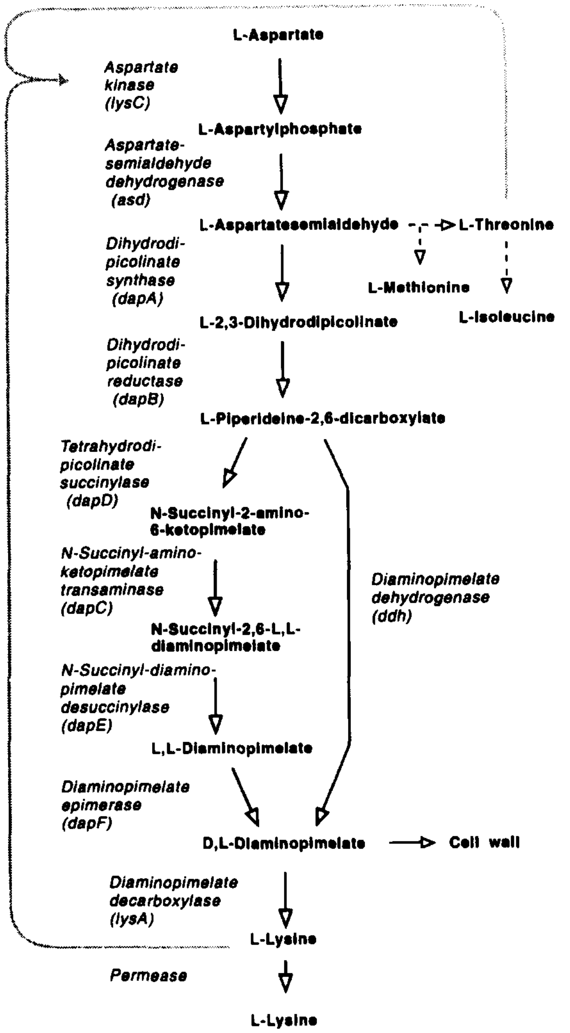

Fig. 1. The split pathway of L-lysine biosynthesis and its regulation in *Corynebacterium glutamicum*. ▬▬▶ = inhibition.

Neither aspartate semialdehyde dehydrogenase, dihydrodipicolinate reductase, diaminopimelate dehydrogenase, nor diaminopimelate decarboxylase overexpression had any effect on lysine overproduction. Most surprisingly, however, overexpression of dehydrodipicolinate synthase also converted the wild-type into a lysine-secreting strain, although not to the same extent as the deregulated aspartate kinase did. This is a strong indication that dihydrodipicolinate synthase is also involved in the flow control of this pathway.

As shown in Fig. 1, the following two enzymes are competing for the intermediate aspartate semialdehyde: the dihydrodipicolinate synthase, which converts this compound lysine, and the homoserine dehydrogenase which converts this intermediate to threonine. Kinetic studies indicate that homoserine dehydrogenase has a much higher affinity for aspartate semialdehyde than dihydrodipicolinate synthase. Increasing the specific activity of the dihydrodipicolinate synthase by overexpressing the gene a greater quantity of aspartate semialdehyde is converted to lysine which is then secreted. The overexpression of dihydrodipicolinate synthase had also a positive effect on very good lysine overproducers. By combined overexpression of aspartate kinase and dihydrodipicolinate synthase lysine secretion could be further increased (10–20%). These results show that of the six enzymes that convert aspartate to lysine, aspartate kinase and dihydrodipicolinate synthase are responsible for metabolic flux control within this biosynthetic pathway. For the construction of strong lysine-producing strains the activities of these enzymes must be increased.

By sequencing the aspartate kinase gene *lysC*, it was found that within the coding region for a polypeptide of 421 amino acids a second in-frame gene is located which codes for a polypeptide of 172 amino acids. This polypeptide is identical to the carboxy terminus of the larger polypeptide. Both genes, termed *lysC*α and *lysC*β, code for a native aspartate kinase protein which is a tetramer of $\alpha_2\,\beta_2$ structure and has a molecular mass of 125.8 kDa [7]. Comparison of the wild-type gene with the gene coding for the deregulated aspartate kinase revealed a single base pair exchange (C to A) in the *lysC*β region resulting in a substitution of serine^{301} in the wild-type enzyme to tyrosine^{301} in the mutant enzyme. Plasmid-bound expression of this mutated *lysC*β region conferred partial feedback resistant aspartate kinase activity to the wild-type of *C. glutamicum* [6]. Therefore, the β-subunit is important for the allosteric control of aspartate kinase.

As shown in Fig. 2 three different pathways of D,L-diaminopimelate and L-lysine synthesis are known in prokaryotes. All bacteria investigated in detail so far appear to utilize only one of the three pathways [8,9]. For example, *Escherichia coli* is using the succinylase variant and in *Bacillus subtilis*

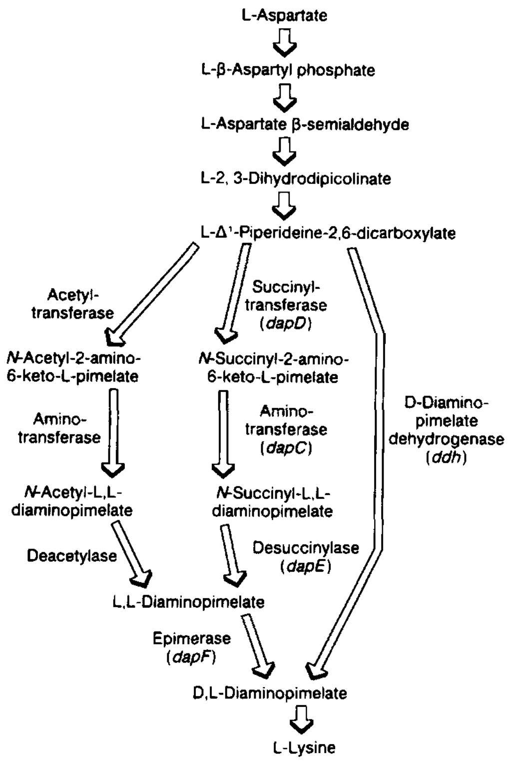

Fig. 2. The three pathways of D,L-diaminopimelate and L-lysine synthesis in prokaryotes.

the acetylase variant is responsible for lysine synthesis. However, recently, Schrumpf et al. [10] have detected that in *C. glutamicum* the dehydrogenase variant and the succinylase variant exist side by side, both allowing D,L-aminopimelate and L-lysine synthesis. This is the only bacterium known so far which has two parallel anabolic pathways, their functions besides the synthesis of D,L-diaminopimelate for cell wall synthesis and lysine synthesis is not known. Mutants with an inactive dehydrogenase pathway are still prototrophic but in overproducers lysine secretion is reduced to 50–70%. Thus, although the dehydrogenase pathway is not essential for growth on mineral salt medium, it is a prerequisite for handling an increased flow of metabolites to diaminopimelate and finally to lysine; for further strain improvement this might be very important.

During the last decade, nuclear magnetic resonance methods have become firmly established as a powerful tool for analysing metabolite concentrations in cell extracts but also in whole cells. Therefore the NMR technique was applied to study the actual physiological flux distribution between the two variants of lysine biosynthesis in the wild-type strain and in several lysine producing strains. By this method the fate of individual carbon atoms can be followed. Sonntag et al. [11] fed cultures with [6-^{13}C]glucose and determined the specific enrichments in carbons of intracellular and excreted lysine and in the metabolite pyruvate which enters the lysine synthetic pathway before it splits into its two variants. This enabled unequivocal determinations of the flux distribution for various strains under a variety of conditions. The dehydrogenase variant was found to contribute about 30% to finally accumutated lysine irrespective of which strain is considered [11,12]. This was unexpected since the specific dehydrogenase activity is always very much higher than that of the succinylase enzymes [10]. Surprisingly, it turned out that the partition coefficient between the two variants varied with cultivation time. At the beginning of the culture the dehydrogenase variant contributed about 70% of the flux, whereas the contribution decreased with cultivation time to nearly zero at the end. This was traced to the availability of ammonium in culture medium. At high ammonium concentrations the dehydrogenase variant carried more than 50% of the total flux whereas in the absence of external amminoum and in the presence of glutamate as sole nitrogen source only the succinylase variant is operative (Table 2). Since the dehydrogenase has a high K_m of 36 mM for NH_4^+ [13], it is conclusive that ammonium availability directly influences the contribution of this enzyme for lysine synthesis by its kinetic characteristics. This example shows the

Table 2
Nitrogen-dependent flux partitioning in *Corynebacterium glutamicum*

Strain	Nitrogen source	Lysine (mM)	Growth OD_{600}	Contribution of dehydrogenase (%)
DG52-5	140 mM Glutamate	4.6	22	0
MH20-22B	600 mM $(NH_4)_2SO_4$	4.3	11	51
MH20-22B	150 mM $(NH_4)_2SO_4$	5.6	14	24
MH20-22B	38 mM $(NH_4)_2SO_4$	3.7	14	6
MH20-22B	23 mM $(NH_4)_2SO_4$	1.9	12	0

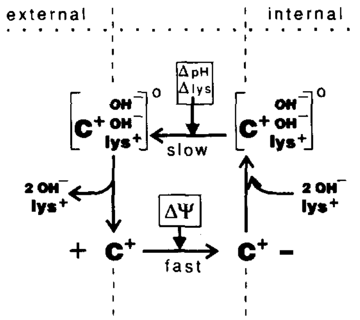

Fig. 3. Kinetic model for lysine export in *Corynebacterium glutamicum*.

extremely powerful use of the NMR technique to determine accurate split ratios of pathways. These techniques will be extended to other main pathways in *C. glutamicum* in order to study the central metabolic network of this organism.

In addition to all steps considered so far to be important for lysine overproduction, the secretion of lysine into the culture medium also has to be noted. Although it is still widely accepted that excretion of amino acids may occur as a result of a physically changed (leaky) membrane [14], recently Krämer and co-workers provided evidence for the presence of specific secretion carriers [15–17]. While in the cells of *C. glutamicum* high concentrations of glutamate (approx. 200 mM) and lysine (approx. 50 mM) could be detected, only lysine was secreted into the culture medium. Thus, the secretion of lysine is not the consequence of unspecific permeability of the plasma membrane but is mediated by a secretion carrier which is specific for lysine. The results of all experimental data are summarized in the model shown in Fig. 3 [17]. In *C. glutamicum*, lysine is excreted in symport with two OH^- ions. The substrate loaded carrier is uncharged. After lysine is transported through the membrane, the carrier changes its orientation back to the cytosol. The pH gradient and the lysine gradient are important for the translocation of the loaded carrier. The membrane potential is important for the reorientation of the positively charged unloaded carrier. All data clearly show that this carrier is a well-designed secretion carrier because (i) it has a rather high K_m value for lysine (20 mM); (ii) it is an OH^- symport system in contrast to the uptake systems, that are H^+ antiport systems; and (iii) it is positively charged, thus the membrane potential stimulates secretion.

Recent analysis of lysine-hyperproducing *C. glutamicum* strains indicates that this secretion carrier has a strong influence on the overproduction of this amino acid [18]. Thus, for the construction of strong lysine-overproducing strains by using the gene cloning technique, the overexpression of the gene(s) for this export system seems also necessary.

3. L-Threonine producing strains

Another example of strain improvement by recombinant DNA techniques is the amplification of the threonine biosynthetic genes in *C. glutamicum*. In *C. glutamicum* the carbon flux from aspartate to threonine is firstly subjected to the control exerted by the regulatory features of aspartate kinase, and further at the level of homoserine dehydrogenase and homoserine kinase (Fig. 4). This enzyme is feedback inhibited by L-threonine (apparent $K_i = 0.16$ mM) and by isoleucine (apparent $K_i = 5$ mM) [5]. In addition, the synthesis of both homoserine dehydrogenase and homoserine kinase was reported to be repressed three-fold by the addition of methionine to the growth medium [19,20]. The other two enzymes involved in threonine biosynthesis, aspartate semialdehyde dehydrogenase and threonine synthase do not seem to be subjected to any inhibition or repression [19,21,22]. The specific activities of the threonine biosynthetic enzymes, determined in cell-free extracts from wild-type *C. glutamicum* (ATCC 13032), and their regulatory features are summerized in Table 3.

Mutants of *C. glutamicum* possessing a homoserine dehydrogenase insensitive to feedback inhibition by L-threonine were obtained by classic mutagenesis procedures and subsequent selection on media containing the threonine analogue 2-amino-3-hydroxyvalerate [22–25]. Recently, we isolated and analysed a *hom* gene (hom_{FBR}) coding for a feedback-resistant homoserine dehydrogenase from a *C. glutamicum* mutant DM 368-3 [22]. Nucleotide sequence analysis and comparison to the wild-type *hom* gene revealed that a single transition from G to A in hom_{FBR} leading to replacement of glycine^{378} by glutamate^{378} in the mutant homoserine dehydrogenase is responsible for deregulation [26]. Using side-directed mutagenesis we replaced the GAG codon for glutamate^{378} by CAG coding for glutamine and by GCG coding for alanine and studied the sensitivity of the altered homoserine dehydrogenase with respect to threonine. As shown in Fig. 5, these changes had a significant effect on the inhibition kinetics. Half maximal inhibition was observed at 0.15 mM L-threonine for the wild-type enzyme (Gly^{378}), at 4 mM for the enzyme possessing Ala^{378}, at 20 mM for the enzyme possessing Gln^{378}, and at approximately 100 mM for the enzyme from *C. glutamicum* DM 368-3 (Glu^{378}). Thus, it is obvious that the size as well as the charge of the side chain of amino acid 378 has an effect on the threonine inhibition of homoserine dehydrogenase. Archer et al. [25] also characterized a *hom* allele (designated *hom*^{dr}) coding for a feedback-resistant homoserine dehydrogenase from an independently isolated *C. glutamicum* mutant R102. This enzyme was 20% inhibited at 50 mM L-threonine. In the *hom*^{dr} gene, a single deletion of a G led to a frameshift in the *hom* reading frame and thus to ten amino acid changes and a deletion of the seven carboxy terminal amino acids relative to the wild-type enzyme. The analysis of both *hom* alleles, *hom*$_{FBR}$ and *hom*^{dr}, revealed the mutation to be located at the carboxy terminus of the homoserine dehydrogenase. Thus, the carboxy terminal part of monofunctional homoserine dehydrogenases may be the domain responsible for feedback inhibition.

Amplification of the feedback inhibition-insensitive homoserine dehydrogenase and homoserine kinase in a high-lysine-overproducing strain permitted channeling of the carbon flow from the intermediate aspartate semialdehyde toward homoserine, resulting in a high accumulation of L-threonine. The final lysine concentration was shifted from 65 g/l to 4 g/l and the final threonine concentration was increased from 0 g/l to 52 g/l (Fig. 6) [27]. Overexpression of the third gene for threonine biosynthesis, threonine synthase alone or in combination with the other two genes had no further effect on threonine or lysine overproduction [28]. However, threonine production could be increased by 12% by the expression of cloned phosphoenolpyruvate carboxylase [29]. This improvement can be explained by the fact that in this strain the formation of the precursor oxaloacetate and thus the carbon flow into the pathway of the biosynthesis of the aspartate family of amino acids is increased.

4. L-Isoleucine-producing strains

Isoleucine is derived from threonine and synthesized in five reactions (Fig. 4) which are catalysed by (i) threonine dehydratase, (ii) acetohydroxy acid

Table 3
Specific activities and regulatory properties of *Corynebacterium glutamicum* enzyme involved in L-threonine and L-isoleucine biosynthesis

Enzyme	Specific activity (μmol min^{-1} (mg protein)$^{-1}$)	Enzyme activity inhibited by [a]	Corresponding gene(s)	Gene expression regulated by [a]
Aspartate kinase	0.010	lys + thr	lysC(lysCα + lysCβ)	–
Aspartate semialdehyde dehydrogenase	0.016	–	asd	–
Homoserine dehydrogenase	0.830	thr + ile	hom	met
Homoserine kinase	0.022	(thr)	thrB	met
Threonine synthase	0.006	–	thrC	–
Threonine dehydratase	0.020	ile	ilvA	–
Acetohydroxy acid synthase	0.030	ile, leu, val	ilvB, ilvN	ile, leu, val (2-oxobutyrate)
Isomeroreductase	0.034	–	ilvC	–
Dihydroxy dehydratase	0.084	–	ilvD	–
Transaminase B	0.120	–	ilvE	–

[a] lys, lysine; thr, threonine; ile, isoleucine; leu, leucine; val, valine; met, methionine

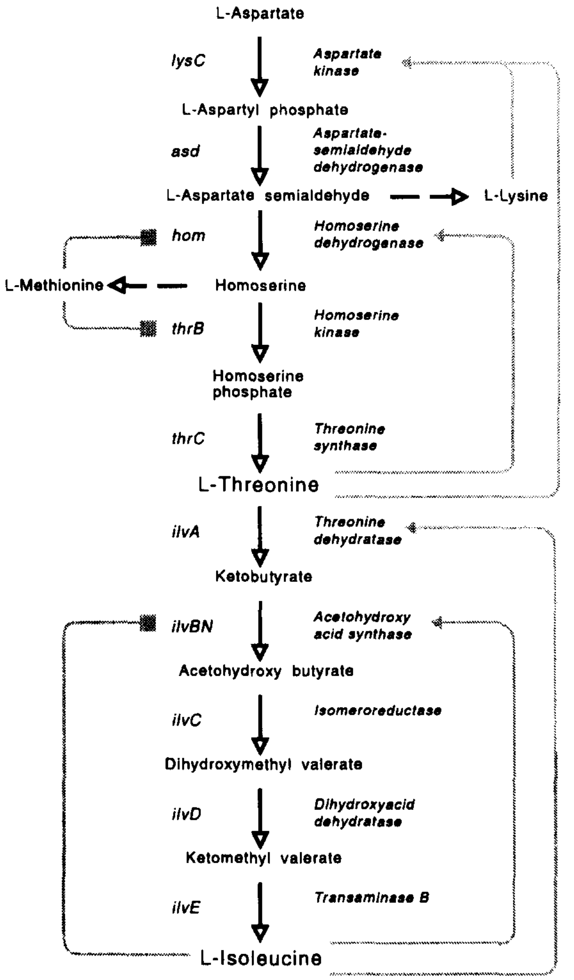

Fig. 4. Biosynthetic pathway of L-threonine and L-soleucine and its regulation in *Corynebacterium glutamicum.* ▬▬▶ = inhibition; ▬▬■ = repression.

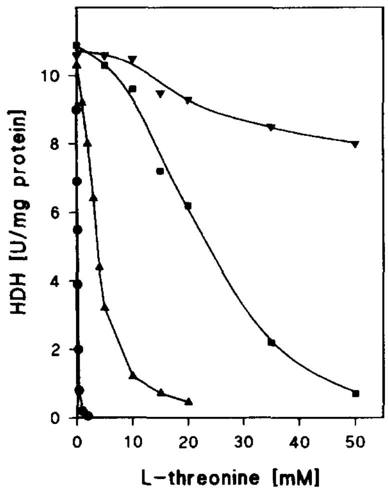

Fig. 5. Threonine-sensitivity of wild-type *Corynebacterium glutamicum* homoserine dehydrogenase (HDH) and various recombinant HDHs changed in amino acid 378. The specific HDH activities were determined in cell-free extracts from *Corynebacterium glutamicum* strains carrying the genes for the respective HDH on plasmid. (●) wild-type HDH-Gly[378]; (▲) HDH-Ala[378]; (■) HDH-Gln[378]; (▼) HDH-Glu[378].

synthase, (iii) isomeroreductase, (iv) dihydroxy acid dehydratase, and (v) transaminase B. Only the first of these enzymes is isoleucine-specific, since the other four are also involved in valine and leucine biosynthesis. In contrast to enterobacteria, in which up to five isoenzymes of acetohydroxy acid synthase have been found [30], *C. glutamicum* possesses only one acetohydroxy acid synthase [31–33].

The carbon flux from threonine to isoleucine is controlled by threonine dehydratase and by acetohydroxy acid synthase. There are no indications of regulation of the last three enzymes involved, neither on the genetic nor the enzyme level. For the *C. glutamicum* threonine dehydratase isoleucine is a negative allosteric effector which increases the apparent K_m value for threonine from 21 to 78 mM and valine acts as a positive allosteric effector reducing the K_m value to 12 mM [34,35]. Thus, with respect to these effectors the *C. glutamicum* enzyme is regulated similarly to that of *E. coli*. The other enzyme important for carbon flux control, acetohy-

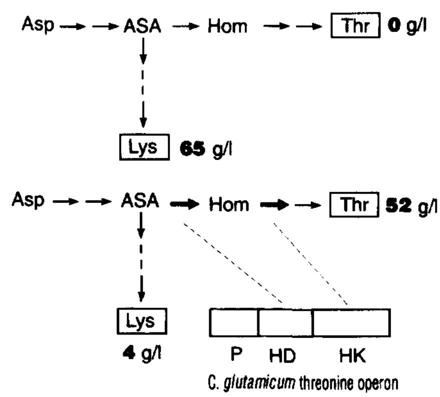

Fig. 6. Diversion of carbon flow by introduction of the *Corynebacterium glutamicum* threonine operon into a lysine-producing strain [27].

droxy acid synthase, is feedback inhibited to approximately 50% by the three branched chain amino acids [31]. This is comparable to the situation found for two of the *E. coli* isoenzymes [30]. The amount of *C. glutamicum* acetohydroxy acid synthase in the cells is strictly regulated. The addition of 2-oxobutyrate results in an up to 20-fold increase of the specific activity [31]. This is assumed to be due to the high affinity of the acetohydroxy acid synthase to 2-oxobutyrate, which might result in a intracellular leucine and valine shortage. Keilhauer et al. [33] recently showed that expression of the acetohydroxy acid synthase genes in *C. glutamicum* is regulated at the transcriptional level.

Although it is evident that the carbon flux from threonine to isoleucine is controlled by the regulation of threonine dehydratase and acetohydroxy acid synthase, up to now there have been no reports on the isolation of *C. glutamicum* mutants possessing one or both of the enzymes in a deregulated fashion. Recently the gene coding for the enzyme threonine dehydratese were cloned from *C. glutamicum* and sequenced [35]. Thus, by site directed mutagenesis, the allosteric domain of the enzyme can be changed to overcome the feedback control. To obtain effective isoleucine overproduction, the mutated strain should also possess a deregulated aspartate kinase and a deregulated homoserine dehydrogenase to allow an appropriate supply of threonine. By the addition of the precursor 2-oxobutyrate to growing cells of *C. glutamicum* and by bypassing the control of acetohydroxy acid synthase by leucine shortage it was possible to obtain an accumulation of up to 19 g/l isoleucine in the medium [31]. This result shows that acetohydroxy acid synthase is not as tightly controlled as aspartate kinase or homoserine dehydrogenase.

Ebbighausen et al. [15] demonstrated a specific isoleucine export carrier which is most probably secondary active. It is not entirely clear whether this carrier limits the isoleucine excretion into the medium. However, with hydroxybutyrate as precursor for isoleucine synthesis Wilhelm et al. [36] found a variety of degradation products of the intermediates of the isoleucine biosynthetic pathway. Since also degradation products of the last intermediate were detected it can be assumed to be present in a rather high cytosolic concentration. A limitation of the isoleucine excretion would increase the internal concentration of isoleucine and its precursors and thus explain this result.

6. Conclusion

In the last years the biochemistry, physiology and molecular biology of amino acid biosynthesis in *C. glutamicum* have been studied intensively and, especially by development and application of genetic techniques, a lot of information has emerged. The current knowledge on the biosyntheses of lysine, threonine and isoleucine and their regulatory features shows that the pathways leading to these amino acids are much simpler in their regulation when compared e.g. to the corresponding pathways in *E. coli*. In *C. glutamicum*, only very few of the involved enzymes are controlled and up to now no isoenzymes could be detected. This may be a reason why this bacterium can be manipulated quite well for the production of various amino acids.

Single and/or combined cloning and expression of the genes or disruption of certain genes in *C. glutamicum* enabled the analysis of carbon flux control in response to elevation or removal of the respective enzyme activity. Based on these analyses, new strategies for the manipulation of this industrially important amino acid producer become possible. So far, the development of highly productive strains was based on random mutagenesis, selection and screening procedures and thus has been an empirical and undefined process. Detailed information on the biosynthetic pathways and their regulation and the availability of the genes involved now permit directed metabolic design, i.e. improvement of amino acid biosynthesis and excretion by manipulation of enzymatic and/or regulatory functions of *C. glutamicum* with the application of recombinant DNA technology.

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